

MADHAV KUMAR YADAV

Data Scientist | AI/ML Python Developer

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PROFESSIONAL SUMMARY

Final-year B.Tech (CSE) student with roots in both cybersecurity and data analytics, now building toward applied machine learning. Have built and deployed 3 end-to-end ML pipelines in Python (88–96% accuracy) covering classification, regression, and security-relevant threat detection – each containerized and shipped as a working service, not left as a notebook. Comfortable with Docker, FastAPI, and MLflow for turning a trained model into something that runs. Cisco coursework covers network defense and threat monitoring; Infosys Springboard and Power BI training cover analytics and business reporting. Looking for an entry-level Data Scientist or ML Engineer role where both backgrounds can be put to use.

TECHNICAL SKILLS

Programming & Querying: Python, SQL, MySQL

ML & Data Libraries: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, SciPy

ML Techniques: Supervised Learning, Classification, Regression, EDA, Feature Engineering, Hyperparameter Tuning, Cross-Validation

MLOps & Deployment: Docker, FastAPI, Flask, Streamlit, MLflow, REST APIs, Render

Data Analytics & BI: Power BI, Power Query, DAX, Microsoft Excel, ER Modeling, Data Visualization

Security Fundamentals: Network Defense, Incident Response, Threat Monitoring, Anomaly Detection

Cloud & Concepts: Cloud Computing Fundamentals, Big Data Concepts, Agile/Scrum

Databases & Tools: MongoDB, MySQL, Git, GitHub, Jupyter Notebook, VS Code

PROJECTS

Network Security – Phishing & Intrusion Detection | GitHub

Jan 2026

Tech Stack: Python, Scikit-learn, Streamlit, MLflow, MongoDB, Docker

- Trained and tuned ML classification models to flag phishing websites and malicious network activity, landing at **96% test accuracy** after several rounds of feature engineering and hyperparameter search – the network-security angle came directly from the Cisco coursework.
- Used **MLflow** to track 50+ experiment runs and **MongoDB** to store results, making it easier to compare model versions side by side instead of losing track of what changed between runs.
- Packaged the pipeline in **Docker** and put a **Streamlit** front end on it so predictions could be run in batches without touching the code.

Student Performance Predictor – End-to-End MLOps Pipeline | GitHub | Live Demo

Dec 2025

Tech Stack: Python, Scikit-learn, FastAPI, Docker, Pandas, MLflow

- Built a scalable MLOps pipeline predicting student outcomes, reaching an **88% F1-score** across 15+ engineered features on 5,000+ records.
- Exposed the model through a **FastAPI** REST endpoint and containerized the service with **Docker** so it could be deployed the same way regardless of environment.
- Cut inference latency by **30%** (150 ms → 105 ms) by profiling the request path and removing unnecessary preprocessing steps, tracked through MLflow.

Forest Fire Weather Index (FWI) Prediction | GitHub

Oct 2025

Tech Stack: Python, Scikit-learn, Flask, Pandas, Render

- Built a Flask web app predicting the Forest Fire Weather Index at **92% accuracy**, using the Algerian Forest Fire dataset (244 instances, 11 features).
- Ran EDA to find the strongest fire-risk drivers, then optimized a Linear Regression model with GridSearchCV and 5-fold cross-validation.
- Deployed on **Render** and wrote up the API so the model could be reused without re-reading the source code.

EDUCATION

Bachelor of Technology in Computer Science

2023 – 2027

Jagannath University Jaipur, Chaksu, Rajasthan, India

Relevant Coursework: Machine Learning, Data Structures & Algorithms, DBMS, Probability & Statistics, Big Data Analytics, Linear Algebra, Calculus, Applied Mathematics

CERTIFICATIONS & TRAINING

- **Cybersecurity Fundamentals** – Cisco Networking Academy (2024–2025): Junior Cybersecurity Analyst Career Path; Introduction to Cybersecurity – network defense, incident response, threat monitoring.
- **Python & Programming** – Cisco Networking Academy (2024–2025): PCAP / Python Essentials 1 – core syntax, data structures, functions, control flow.
- **Data Analytics & Visualization** – Infosys Springboard & Udemy (2026): Power BI Foundation Certification; ER Modeling, Excel, Data Science Foundations; Power Query & DAX.
- **Cloud, Big Data & Agile** – Infosys Springboard (Apr 2026): Cloud Computing Fundamentals, Big Data Concepts, Agile/Scrum.